Qi Wu

Optical Characterization & Metrology | Spectroscopy · Simulation · Instrument Development

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Profile. Optical characterization and metrology researcher with four years at imec developing optical methods to characterize semiconductor surfaces and nano-scale structures. Experience spans infrared spectroscopy, optical simulation, and the design and building of custom optical measurement systems. Cross-disciplinary background across optics, electrical engineering, and MEMS. Strong in optical data analysis in Python and MATLAB.

EDUCATION

Ph.D. in Chemistry (optical metrology)	KU Leuven & imec, Leuven, Belgium | 2022–2026 (expected)
M.Sc. in Electrical Engineering	University of Twente, the Netherlands | 2019–2021
B.Eng. in Electrical Engineering	Northwestern Polytechnical University, China | 2015–2019

TECHNICAL SKILLS

Optical Metrology & Spectroscopy: ATR-FTIR spectroscopy for in-situ interfacial and surface analysis; ellipsometry; dynamic contact-angle / wettability analysis; in-line optical monitoring.

Optical Simulation & Modelling: FDTD simulation of IR, evanescent-field, and near-field optics; total-internal-reflection modelling; mapping optical signals to physical surface states.

Optical Instrumentation: Design, build, and re-engineering of custom optical measurement systems and modules; optical alignment; embedded control; rapid prototyping (3D printing); measurement automation.

Programming & Data: Python and MATLAB for spectral fitting, error analysis, data processing, and hardware-software integration.

Surface & Cleanroom: Surface functionalization (SAMs), thin-film CVD; SEM, AFM; 2,000+ hrs cleanroom.

Languages: English (professional), Dutch (beginner), Mandarin Chinese (native).

RESEARCH EXPERIENCE

PhD Researcher — Optical Metrology & Instrumentation *KU Leuven & imec, Leuven, Belgium | 2022–Present*

- **Custom optical measurement system:** Designed and built a specialized ATR-FTIR module with adjustable optics, re-engineering a commercial instrument to enable in-situ optical measurements not possible with the stock setup.
- **Optical simulation (FDTD):** Built an FDTD framework modelling broadband IR under total internal reflection in 3D silicon; simulated how sub-wavelength surface structures reshape the near-field, and mapped optical signals to physical surface states.
- **Optical characterization of nanostructures:** Characterized nano-patterned surfaces by optical methods, SEM, and AFM, including geometry dependence and deviations from theoretical models; built a dual-axis imaging module for anisotropic wetting analysis.
- **Defect inspection (SEM):** Used SEM to inspect pattern collapse in high-aspect-ratio structures and linked the defect to a specific process parameter (solvent surface tension).
- **Process & data:** Established a robust process window with >90% run-to-run reproducibility; automated spectral fitting and analysis in Python and MATLAB.

Graduate Intern *ARCNL, Amsterdam, the Netherlands | 2021*

- **Simulation & collaboration:** Stress-strain simulations (COMSOL) of elastomer microstructures for low-wear wafer carriers; worked with cleanroom specialists in a research-to-manufacturing setting.

PUBLICATIONS & PRESENTATIONS

- Q. Wu et al., *"Modulating Silane Monolayer Grafting on Nanopatterned Silicon: Insights from Infrared-Spectroscopy-Based Air-Trapping Monitoring,"* Solid State Phenomena (UCPSS 2025), accepted / in press. **Best Student Paper Award.**
- Q. Wu, *"Impact of HF Pretreatment on Silane Self-Assembled Monolayers on Silicon Surfaces,"* oral presentation, 18th Mini-Symposium on Liquids, Okayama University (online).
- Q. Wu, *"Optical Method to Characterize the Wetting States on Nano-patterned Surfaces,"* poster, SPP 2171 Summer School.

INTERESTS

Tennis, bouldering, hiking, DIY electronics.